```
    if i<nc then printf(" & ");
    end
end
printf(" \\\\"+"\n"+"\\hline"+"\n");
// Data values
for i=2:nr
    for j=1:nc
        printf("%"+string(w(j))+"s",data(i,j));
        if j<nc then printf(" & ");
        end
    end
    printf(" \\\\"+"\n");
end
printf(+msprintf("\\\\end{tabular}")+"\n"..
      +msprintf("\\\\end{table}")+"\n");
```

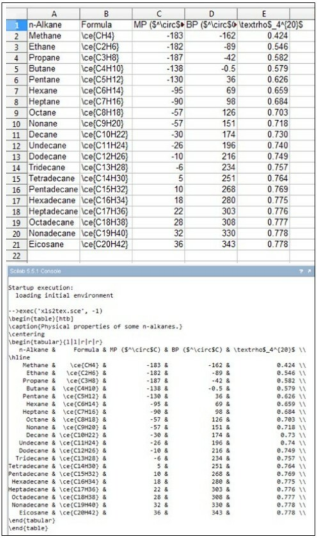
Keywords

The keywords of Scilab fall into five categories: *primitives*, *commands*, *variables*, *functions* and *xcos functions*. In my Scilab installation, five toolboxes — GUI Builder, IPT3, JSON, NaN and Quapro — are present, apart from some other dependencies that got automatically installed. The function *getscilabkeywords* returns a list with all the Scilab keywords already categorised.

```
clear;
list_all=getscilabkeywords();
write("1_primitives.txt",list_all(1));
write("2_commands.txt",list_all(2));
write("3_variables.txt",list_all(3));
write("4_functions.txt",list_all(4));
write("5_xcosfuns.txt",list_all(5));
```

Now I have five text files. But there is a small problem. Some keywords are present in more than one file - the keywords extraction is not really perfect, so some deletions are necessary. The following keywords are removed from the *functions* file:

```
// Also present in primitives
datatipManagerMode
datatipMove
datatipSetOrientation
datatipSetStyle
// Also present in commands
apropos
help
// Also present in xcosfuns
lincos
scicos_simulate
steadycos
block_parameter_error
find_scicos_version
fixedpointgcd
```



	A	B	C	D	E
1	n-Alkane	Formula	MP ($^\circ$	BP ($^\circ$	\textrho$_4^{20}$
2	Methane	\ce{CH4}	-183	-162	0.424
3	Ethane	\ce{C2H6}	-182	-89	0.546
4	Propane	\ce{C3H8}	-187	-42	0.582
5	Butane	\ce{C4H10}	-138	-0.5	0.579
6	Pentane	\ce{C5H12}	-130	36	0.626
7	Hexane	\ce{C6H14}	-95	69	0.659
8	Heptane	\ce{C7H16}	-90	98	0.684
9	Octane	\ce{C8H18}	-57	126	0.703
10	Nonane	\ce{C9H20}	-57	151	0.718
11	Decane	\ce{C10H22}	-30	174	0.730
12	Undecane	\ce{C11H24}	-26	196	0.740
13	Dodecane	\ce{C12H26}	-10	216	0.749
14	Tridecane	\ce{C13H28}	-6	234	0.757
15	Tetradecane	\ce{C14H30}	5	251	0.764
16	Pentadecane	\ce{C15H32}	10	268	0.769
17	Hexadecane	\ce{C16H34}	18	280	0.775
18	Heptadecane	\ce{C17H36}	22	303	0.776
19	Octadecane	\ce{C18H38}	28	308	0.777
20	Nonadecane	\ce{C19H40}	32	330	0.778
21	Eicosane	\ce{C20H42}	36	343	0.778
22					

```
Scilab 5.5.1 Console

Startup execution:
  loading initial environment

-->exec('xls2tex.sce', -1)
\begin{table}[htb]
\caption{Physical properties of some n-alkanes.}
\centering
\begin{tabular}{l|l|r|r|r}
   n-Alkane &     Formula & MP ($^\circ$C) & BP ($^\circ$C) & \textrho$_4^{20}$ \\
\hline
    Methane &    \ce{CH4} &           -183 &           -162 &             0.424 \\
     Ethane &   \ce{C2H6} &           -182 &            -89 &             0.546 \\
    Propane &   \ce{C3H8} &           -187 &            -42 &             0.582 \\
     Butane &  \ce{C4H10} &           -138 &           -0.5 &             0.579 \\
    Pentane &  \ce{C5H12} &           -130 &             36 &             0.626 \\
     Hexane &  \ce{C6H14} &            -95 &             69 &             0.659 \\
    Heptane &  \ce{C7H16} &            -90 &             98 &             0.684 \\
     Octane &  \ce{C8H18} &            -57 &            126 &             0.703 \\
     Nonane &  \ce{C9H20} &            -57 &            151 &             0.718 \\
     Decane & \ce{C10H22} &            -30 &            174 &              0.73 \\
   Undecane & \ce{C11H24} &            -26 &            196 &              0.74 \\
   Dodecane & \ce{C12H26} &            -10 &            216 &             0.749 \\
  Tridecane & \ce{C13H28} &             -6 &            234 &             0.757 \\
Tetradecane & \ce{C14H30} &              5 &            251 &             0.764 \\
Pentadecane & \ce{C15H32} &             10 &            268 &             0.769 \\
 Hexadecane & \ce{C16H34} &             18 &            280 &             0.775 \\
Heptadecane & \ce{C17H36} &             22 &            303 &             0.776 \\
 Octadecane & \ce{C18H38} &             28 &            308 &             0.777 \\
 Nonadecane & \ce{C19H40} &             32 &            330 &             0.778 \\
   Eicosane & \ce{C20H42} &             36 &            343 &             0.778 \\
\end{tabular}
\end{table}
```

Figure 4: A table input (xls, top) and output (LaTeX, bottom)

```
get_scicos_version
initial_scicos_tables
returntoscilab
scicos_getvalue
scicos_workspace_init
with_modelica_compiler
```

Finally, I have five text files without any overlap: *primitives* with 1193 keywords, *commands* with 29 keywords, *variables* with 94 keywords, *functions* with 2462 keywords and *xcosfuns* with 12 keywords. The total is equal to 3790 keywords.

GNU Emacs

The keywords in each text file are sorted first by length and then alphabetically. Then it's necessary to add the quotes before and after each keyword, merge every four lines (to reduce the number of rows) and add some Emacs Lisp code. A short example is given by the *commands*: